

KramaBench

A Benchmark for AI Systems on Data- Intensive Tasks

*yes, it's KRama not kaRma

Gerardo Vitagliano, Eugenie Lai, Ziyu
Zhang, Om Chabra, Mike Cafarella, Tim
Kraska, Sam Madden

(*et al*)

NEDB 2026



Data Discovery



Table Extraction



Data Cleaning



Missing Value Imputation



Data Integration



Entity Matching



Statistical Computation



Modeling and Regression

Real-world data science

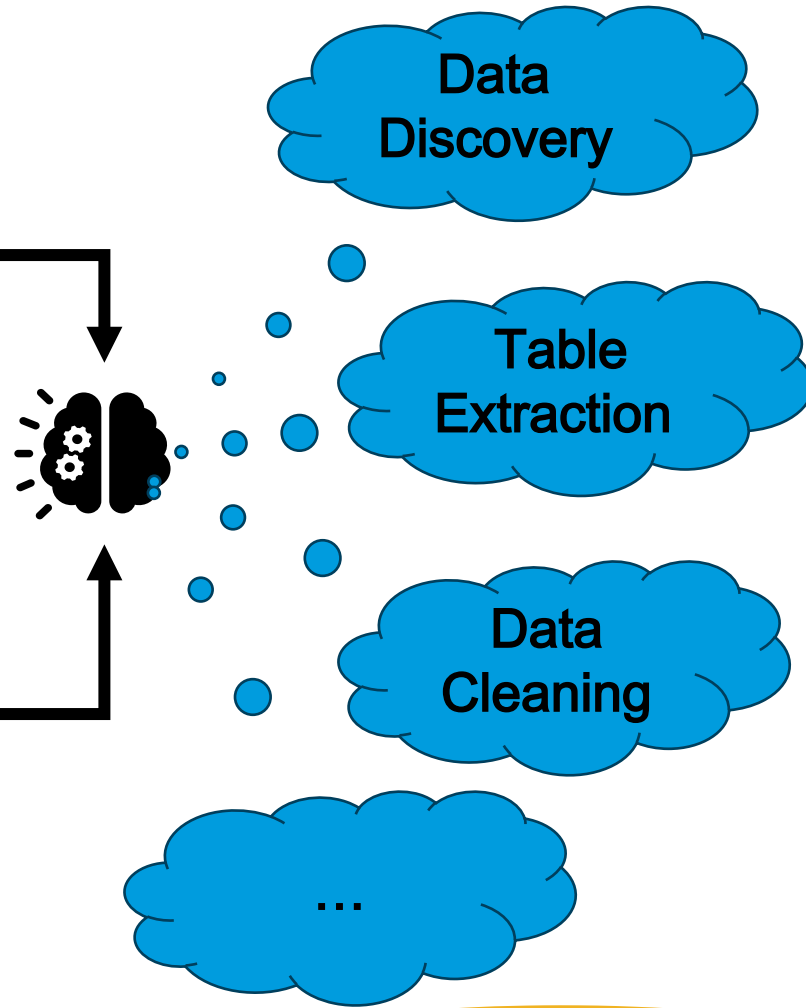
Task According to the Consumer Sentinel Network, what is the total amount of money defrauded in 2024, summing over all payment methods? Give an integer in millions of dollars.

Municipalities
 Albany, NY
 -

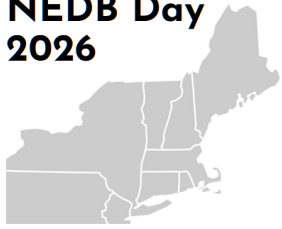
Txn | Payee | Payment Method
 001 | Alice | Card
 002 | Bob. | Credit Card
 * These transactions are from 2024

Consumer Sentinel Network is an agency that...

Fraud By Municipalities
Municipality | Fraud Type
 Albany, NY | Identity Theft

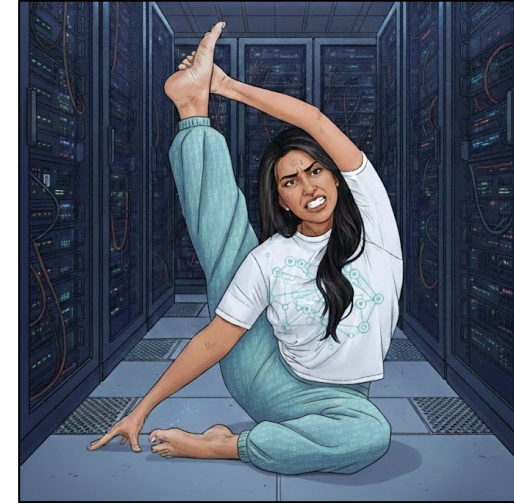


535M



The Benchmark Gap

- Real-world data science is **complex**:
 - Data sources are multimodal, noisy, ambiguous
 - Domain knowledge might be required
 - It requires whole processing pipelines
- But popular benchmarks evaluate **isolated tasks**
 - *Ex:* Data Discovery, Coding, NL 2 SQL, ...
- Our goal is to have an encompassing benchmark!



KramaBench Overview

- Goal: evaluate **automation** of data science pipelines
- We collected data from 6 challenging domains
 - >100 pipelines reproducing published studies
 - Each pipeline is broken into subtasks
 - The tasks span ~1700 real world data files
- We provide reference solutions
 - Verified by external contributors
 - Annotated into key functionalities/subtasks
 - Enable evaluation of planning vs. execution

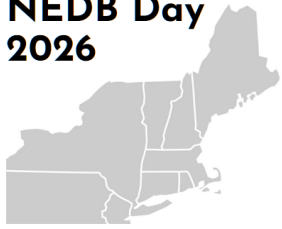
Domain	# tasks	# subtasks	# files	File Size
Archeology	12	71	5	7.5 MB
Astronomy	12	68	1556	486MB
Biomedical	9	38	7	175 MB
Environment	20	148	37	31 MB
Legal	30	188	136	1.3 MB
Wildfire	21	120	23	1 GB
Overall	104	633	1764	1.7GB

Evaluation modalities

- KramaBench provides 3 different scores:
- **End-to-end answer correctness**
 - The answer for each task is scored from $[0,1]$
 - The main score is the average score across all tasks
- **High-level pipeline design**
 - Coverage of required key functionalities
 - Uses LLM-as-a-judge for semantic matching
- **Implementation of pipeline code**
 - Systems are asked to implement fine-grained subtasks
 - We score intermediate outputs

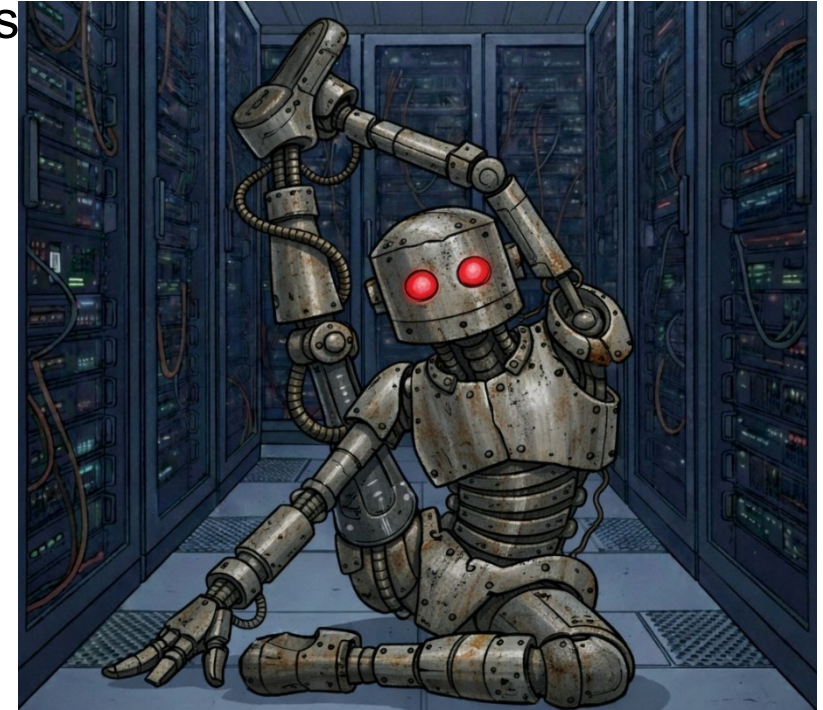
Systems Under Test

- We tested three families of systems with multiple LLM backends
 - GPT family (o3, 4o), Claude 3.5, Llama3.3 DeepseekR1

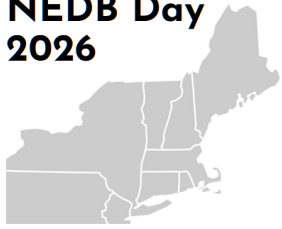


Systems Under Test

- We tested three families of systems with multiple LLM backends
 - GPT family (o3, 4o), Claude 3.5, Llama3.3 DeepseekR1
- **Naïve single-agent baselines**
 - One-shot pass
 - Sampled contents of the data lake

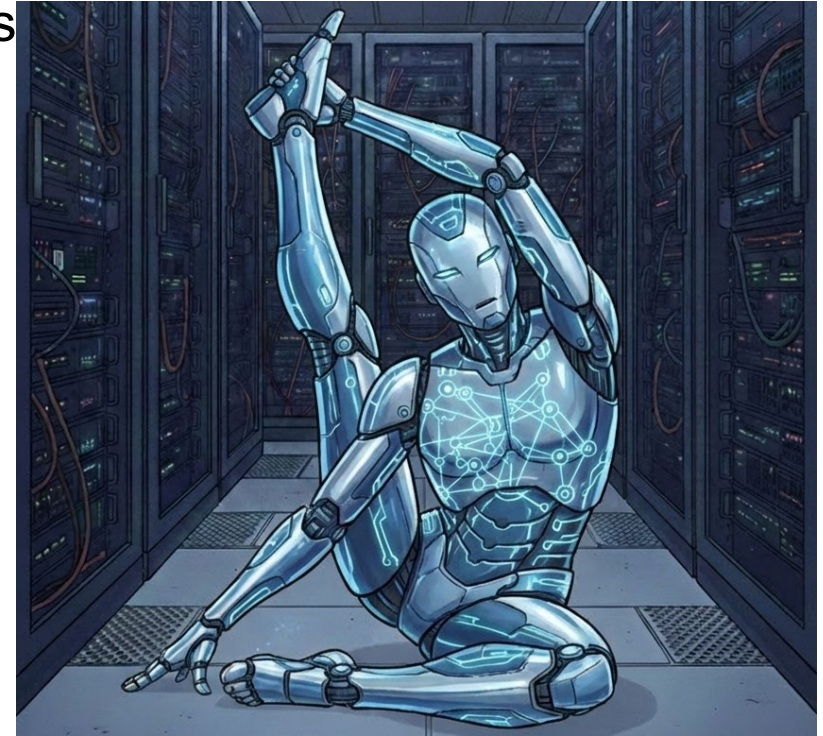


DS-GURU

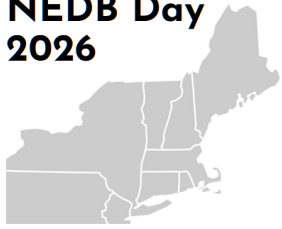


Systems Under Test

- We tested three families of systems with multiple LLM backends
 - GPT family (o3, 4o), Claude 3.5, Llama3.3 DeepseekR1
- Naïve singleagent baselines
 - One-shot pass
 - Sampled contents of the data lake
- **Robust single-agent baselines**
 - Few-shot debugging/iterating
 - Full contents of data lake

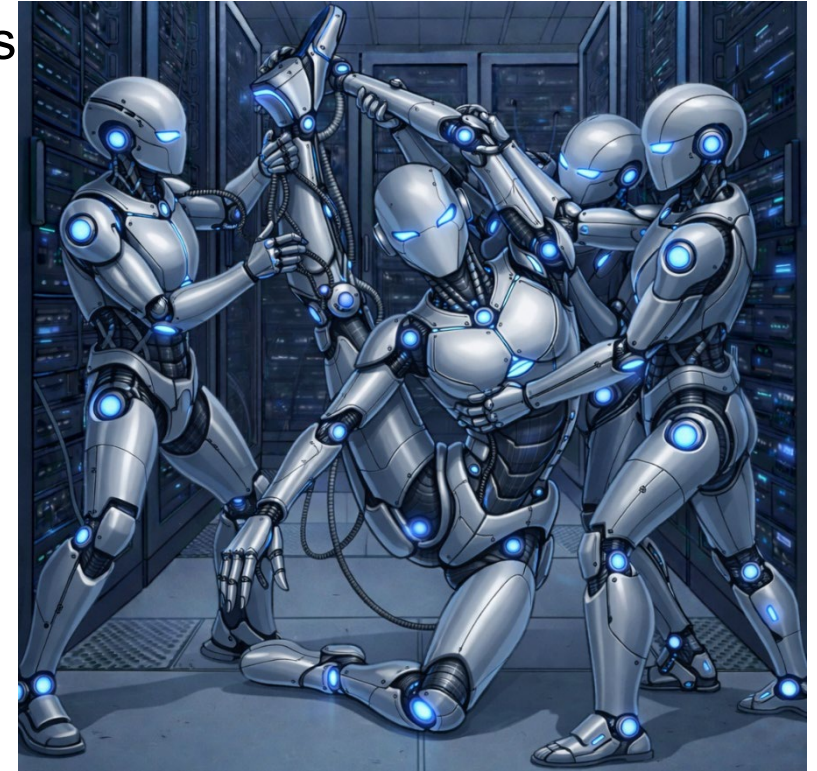


DS-GURU II



Systems Under Test

- We tested three families of systems with multiple LLM backends
 - GPT family (o3, 4o), Claude 3.5, Llama3.3 DeepseekR1
- Naïve singleagent baselines
 - One-shot pass
 - Sampled contents of the data lake
- Robust singleagent baselines
 - Few-shot debugging/iterating
 - Full contents of data lake
- **Deep Research and Multi-agent baselines**
 - Reflection and planning loops
 - Access to web search and tools



smolagentsReflexion

Pipeline design easier than implementation

- Best system **only obtains 52.2% end-to-end accuracy**
- Single-shot LLM fail on most tasks
- Few-shot show improvements
- Multi-agent reflection approaches SOTA
- Pipeline design is easier than implementation
 - Models identify required steps but can't solve them
 - Domain-specific knowledge, e.g., data cleaning

Baseline	End-to-End Acc.	Pipeline Design	Sub-task coding	Avg. Runtime / Task
DS-Guru	9.64%	40.60%	12.95%	fill num
DS-Guru II	22.08%	41.58%	19.75%	0.76 min
smolagents	50.64%	42.14%*	14.23%	6.10 min
OpenAI Deep Research	52.18%	—	—	10.35 min
Human Experts	71.07%	64.00%	—	4.36 min

Domain insights and error analysis

- Common pitfalls
 - Missing joins and multifile aggregations
 - Scientific formats and large files
 - Domain-specific cleaning/preparation
- We include additional ablation experiments
 - Effect of data sampling
 - Effect of perfect retrieval
 - Obscuring data

Introducing KramaBench

- KramaBench features:
 - A benchmark of end-to-end pipelines
 - Real-world data and problems
 - Fine-grained evaluation
- Current AI systems are far from solving it
 - SOTA by Deep Research Agents
 - Pipeline implementation still brittle

Current Rankings

Select Domain: Overall OFF

Search:

RANK	SYSTEM	MODEL	SCORE (%)
1	openAI	Deep Research*	52.2%
2	smolagents DR	Claude-3-7	50.0%
3	smolagents DR	GPT-o3	41.4%
4	smolagents DR	Claude-3-5	41.4%



KramaBench

A Benchmark for AI Systems on Data- Intensive Tasks

Gerardo Vitagliano, Eugenie Lai, Ziyu
Zhang, Om Chabra, Mike Cafarella, Tim
Kraska, Sam Madden

(*et al*)

NEDB 2026

